MODULE 2

Crime Scene Investigation Evidence

Certified Cyber Criminologist - Police / LEA Training

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What You Will Learn

This module is about the **hands of the investigator** - what you do at the scene decides whether the case survives in court.

- **Four forms** of digital evidence - and why each one is dangerously fragile
- **The five-stage framework** - the professional method from identification to chain of custody
- **Seizing a mobile phone** - the six-step checklist you will repeat in every case
- **Hash value** - the digital seal that proves nobody touched the evidence
- **BSA certificate** - the paperwork the court needs before it trusts your evidence

The Opening Story: One Wrong Click

An officer seized a laptop and switched it on **"just to look"**. In court the defence asked one question: **"How do you prove YOU did not create these files?"**

- Switching on the laptop changed file dates and created new logs - the officer could not explain them
- A strong case turned weak. Not because evidence was missing, but because it was **handled wrong**
- **Lesson:** finding evidence is easy. Lifting it correctly is the real investigation

Why Digital Evidence Is So Fragile

Physical evidence waits for you. Digital evidence **can vanish in one second** - sometimes with one wrong click of your own.

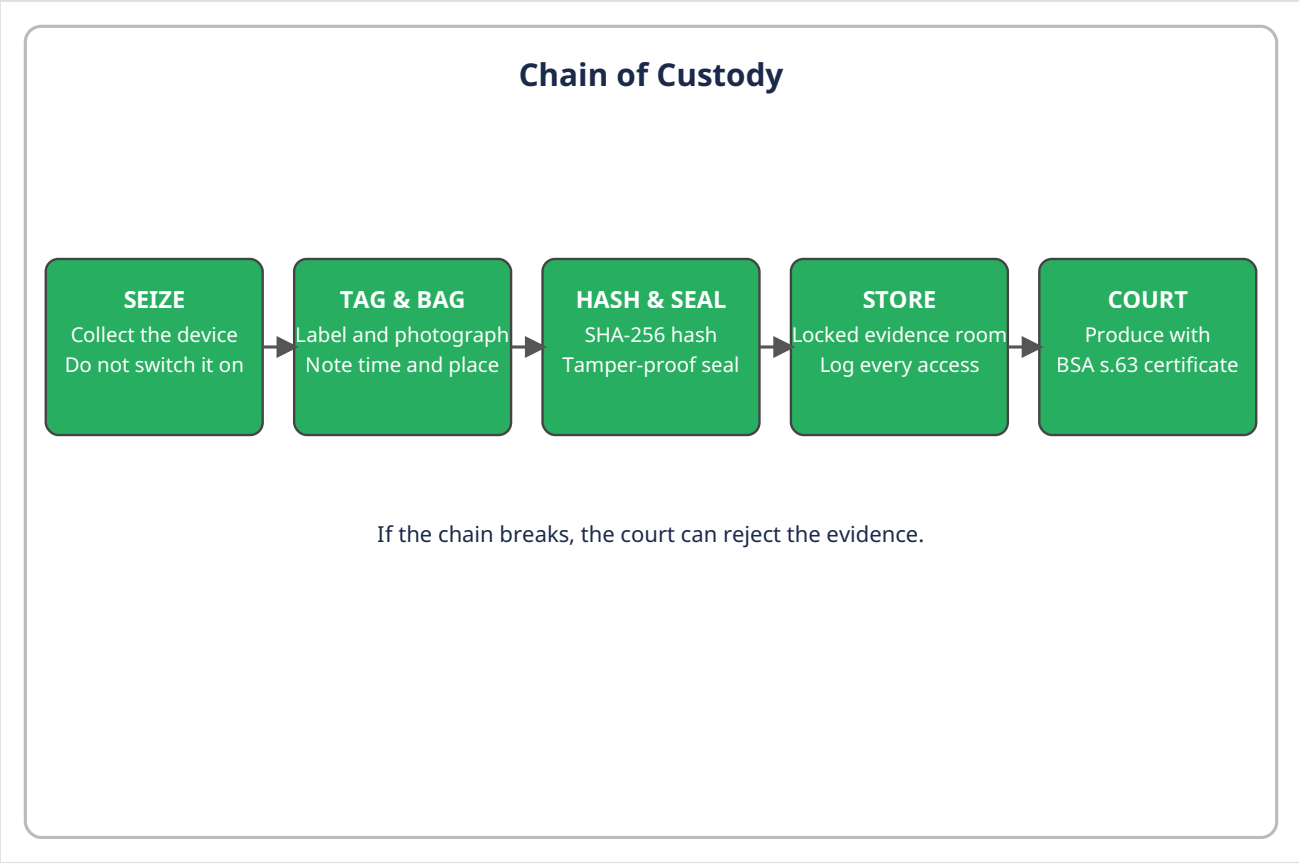
- **Remote wipe:** a seized phone on the network can be erased by the criminal from anywhere, in seconds
- **Timestamps change:** switching on a device rewrites file dates - the defence will call it tampering
- **Volatile data dies:** RAM contents and live connections disappear the moment power goes off
- **Rule:** think before touching. Photograph first, isolate second, seize third

Four Forms of Digital Evidence

Evidence hides in four places. Miss one, and you may miss the **whole story of the crime**.

- **Devices** - phones, laptops, hard disks, pen drives. These are the boxes; the treasure is inside
- **Data** - photos, chats, emails, call records, browsing history. This tells the story
- **Network records** - IP addresses, login times, location data. This tells who was where, when
- **Documents** - bank statements, screenshots, complaint copies. Paper still carries weight in court

Physical vs Digital Evidence



- Digital evidence is **invisible** - you cannot pick it up, you can only copy it
- Its genuineness is proved by **hash value**, not fingerprints
- In court you present the **copy plus certificate** - the original stays sealed

The Five-Stage Framework

Stage	What you do at the scene
1. Identification	Decide what could be evidence - look around, do not hurry
2. Securing	Flight mode or switch off - stop remote wiping and changes
3. Recording	Photograph everything - position, condition, screen
4. Packing	Separate sealed bags, seizure memo written at the spot
5. Chain of custody	Register entry on EVERY change of hands, no blanks

Stage 1: Identification - Look Before You Lift

You cannot seize what you did not notice. Identification means **deciding what could be evidence** before touching anything.

- Scan the whole scene slowly: the phone on the table, the laptop bag, the pen drive in a pocket
- Check **hiding spots** - memory cards, SIM trays, USB drives tucked inside drawers
- Note the **condition** of each item now - ON or OFF, screen locked or open, damage
- Resist the urge to pick things up. **Photograph first** - lifting comes later

Stage 2: Securing - Freeze the Evidence

Securing means putting every device in a state where **its data cannot change** - not by you, not by the criminal.

- **Phone:** flight mode ON, or switch it off. This blocks remote wipe commands instantly
- **Laptop running:** do NOT close the lid. Photograph the screen first - what is open is evidence
- **Do not charge, do not connect,** do not 'just check one message' - every action writes new data
- If a device is already off, **keep it off**. The lab will handle it correctly

Stage 3: Recording - The Camera Is Your Witness

Months later, nobody will remember where the phone was lying. Your **photographs will** - if you take them right.

- Photograph each device **in place** before moving it - position, surroundings, condition
- Capture the **screen** if it is on - open apps and messages are evidence of the moment
- Avoid flash glare on screens and glass - shoot at an angle so text stays readable
- Every photo gets a **written log**: number, date, time, photographer, direction, description

Three Levels of Crime Scene Photography



- **Overall:** the whole room - proves where the scene was
- **Mid-range:** the table or area - shows each item's position
- **Close-up:** the device itself - with a coin or scale for size

The Photo Log: Five Things Per Photo

A photo without a log is just a picture. A photo **with a log is evidence**.

- Write the log **immediately after each photo** - never 'in the evening'. Order goes wrong
- Five fields, every time: **photo number, date, time, who took it, direction** it was taken from
- Add one line of **description**: what is visible, e.g. 'phone mid-table, lock screen on'
- If asked 'when was photo 2 taken?' you must answer **in seconds** - from the log

Stage 4: Packing and the Seizure Memo

Packing protects the evidence physically. The **seizure memo protects it legally** - it is the receipt of the seizure.

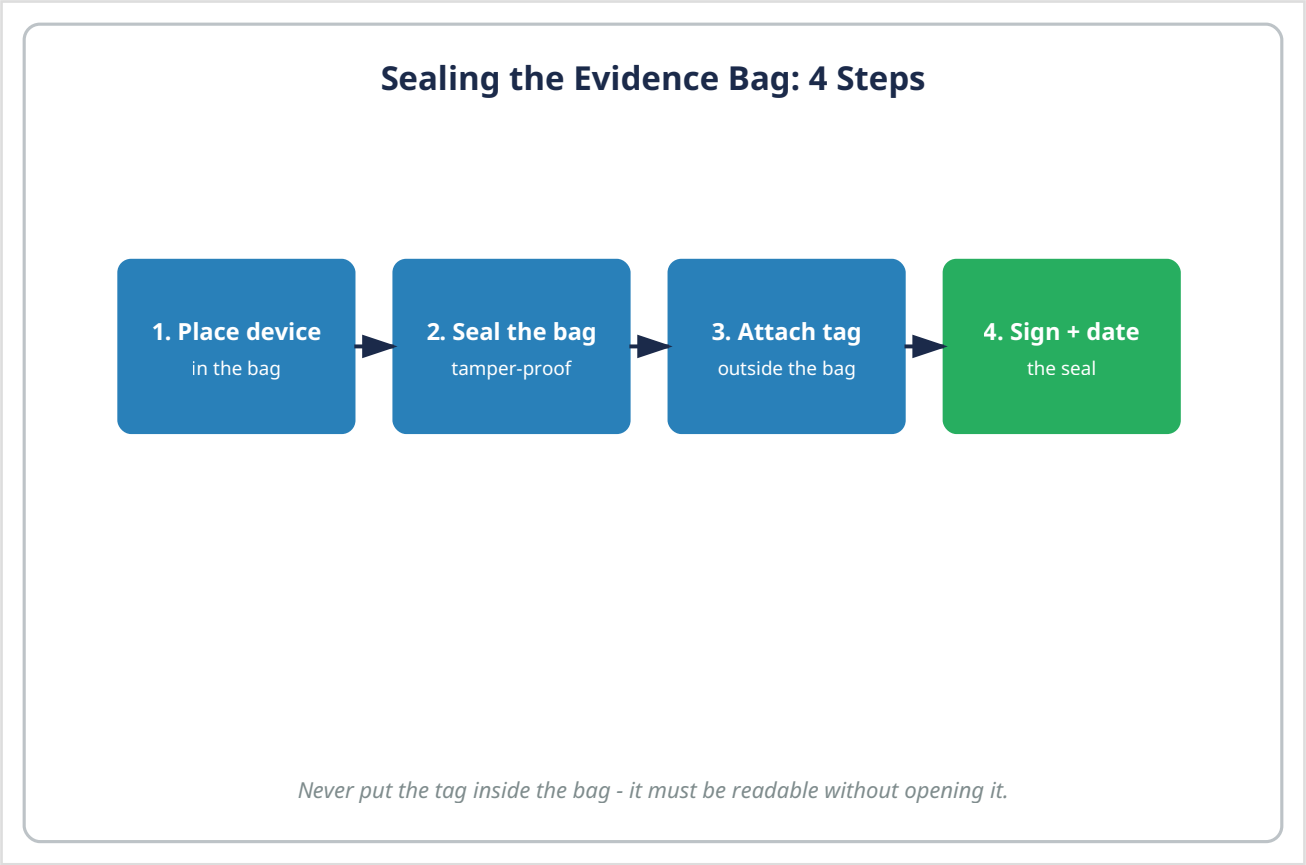
- One device per bag. **Anti-static bag** is best for phones; clean plastic if nothing else
- Label OUTSIDE the bag: case number, date, time, place, your name and signature
- Write the memo **at the spot, not later** - memory fades, courts notice
- Give **one copy to the witness**. The original goes into the case file

Evidence Tag: Every Field Matters



- **No field blank** - case number, sections, date, time, place, description, seal, signatures
- Describe so clearly that **any stranger** reading it can identify the item
- The tag stays **outside** the bag - readable without breaking the seal

Sealing the Evidence Bag



- Seal the bag and write the **seal number** on both the bag and the register
- The seal number is the **link** between the physical bag and the paper trail
- A broken or mismatched seal in court means **instant doubt** on the evidence

Mobile Phone Seizure: Steps 1 to 3

The phone is the most common digital evidence. These six steps must become **muscle memory**.

- **Step 1 - Note the condition:** ON or OFF? What is on the screen? Photograph it before touching
- **Step 2 - Isolate from the network:** flight mode ON or switch off. A connected phone can be wiped remotely
- **Step 3 - Pack with minimal touching:** hold by the edges, anti-static bag is best, avoid pressing buttons

Mobile Phone Seizure: Steps 4 to 6

The last three steps are about **paperwork** - and paperwork is what the court reads.

- **Step 4 - Label the bag:** case number, date, time, place, your name and signature - on the outside
- **Step 5 - Two witnesses:** their names, addresses and signatures on the seizure memo
- **Step 6 - Malkhana deposit:** register entry immediately. Evidence never stays in your pocket or at home

The Seizure Memo: What It Must Contain

The seizure memo is the **legal birth certificate** of the evidence. Seven things, no shortcuts.

- **Case number and sections** - e.g. BNS 318(4), IT Act 66D. Without these the memo floats free
- **Date, time, place** of seizure, plus a **separate description of every item** - colour, condition, where it was found lying
- **Seizing officer's** name, designation, signature - and the **names, addresses, signatures of two witnesses**
- Sample opening: 'Today, 14-10-2026, at 11:30 hours, from room no. 3, in case MOCK/2026/01, in front of witnesses, the following items were seized...'

Stage 5: Chain of Custody

Chain of custody is the **complete life story of the evidence** - whose hands it was in, when, and why. From seizure to court.

- One register page per item. Columns: **date-time, handed by, received by, reason, signatures**
- Three rules: **no blank entries**, no handover without an entry, register stays safe in the malkhana
- Evidence lives in the **malkhana or an authorised place** - never in a pocket, home or private locker
- In court, the defence attacks **this register first**. Complete entries end the attack there

Chain of Custody: A Sample Register

Date and time	Handed by	Received by	Reason
14-10, 11:30	Sub-inspector (you)	Malkhana clerk	Deposited after seizure
16-10, 10:00	Malkhana clerk	Forensic expert	For examination
20-10, 15:00	Forensic expert	Malkhana clerk	Examination complete

When the Chain Breaks: Real Consequences

A gap in the register is a **gift to the defence lawyer**. Here is exactly how it plays out in court.

- 'Sir, where was this phone between the 16th and the 20th?' - **silence** means doubt begins
- 'The phone could have been tampered with during those two weeks' - and the court **will listen**
- One missing signature can make months of technical work **worthless as evidence**
- The register is boring paperwork. It is also the **strongest shield** your case has

Volatile Evidence: Save It First

Volatile evidence **dies in moments**. What is running in RAM, which network connections are open - gone the instant power goes off.

- At the scene: do **not** immediately switch off a running computer. Photograph the screen first
- What is open on the screen **is itself evidence** - chats, files, connections visible right now
- RAM data needs a trained expert - this is called **live forensics**. Call for help, do not improvise
- Hard disk data can wait. **Volatile data cannot**. Secure it first, always

Four Types of Evidence and Their Rules

Type	Examples	Your rule
Volatile	RAM data, live connections	**Secure FIRST** - gone when power goes
Non-volatile	Hard disk, pen drive	Durable, but still handle correctly
Logs	Server and network records	Get them fast - companies delete old logs
Cloud	Drive, iCloud, WhatsApp backup	Needs legal procedure to obtain

Hash Value: The Digital Seal

On paper, a seal proves genuineness. In the digital world, the **hash value** does that job - and it cannot be faked.

- **SHA-256** reads any file and produces a fixed 64-character code - its digital fingerprint
- Change **even one character** in the file and the entire hash changes completely
- Note the hash, and later anyone can verify: **same hash = untouched, different hash = tampered**
- In court, the hash proves the file presented today is the **same file seized on day one**

Three Points to Note the Hash

One hash is good. **Three matching hashes** at three stages make the chain unbreakable.

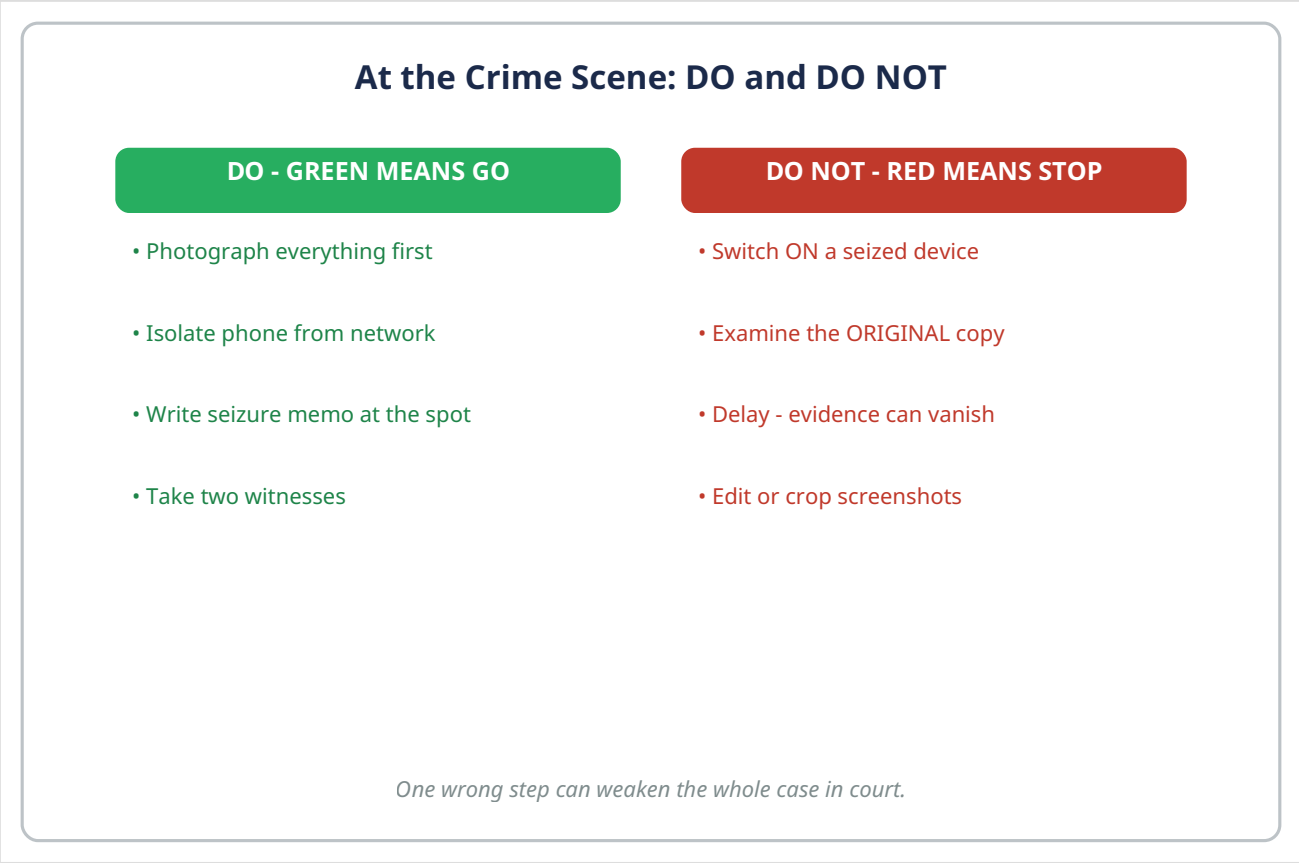
- **At seizure:** if a file is taken separately from the device, note its hash at the spot itself
- **During imaging:** when the forensic copy is made, tools like FTK Imager extract the hash - write it in the case file
- **Before court:** extract the hash again and match. Same hash? Write: 'no change has been made in the evidence'

Write Blocker: The Evidence Protector

Connecting a disk to a computer can **silently write** to it - changing dates, creating files. A write blocker stops that.

- A **hardware write blocker** sits between the disk and the computer - it allows only reading, never writing
- Result: you make an **exact forensic copy** (imaging) with zero changes to the original
- No hardware? **FTK Imager** (free) gives software write-protection while imaging - Lab 4 covers it
- **Golden rule:** the original is never examined. All analysis happens on the forensic copy

DOs and DON'Ts at the Crime Scene



- **DO:** photograph, isolate, label, witness, register - in that order
- **DON'T:** switch on, browse, charge, or delay
- When in doubt, **do nothing** and call the expert

BSA Certificate: What the Court Needs

Under the **Bharatiya Sakshya Adhiniyam (BSA)**, digital records are evidence - but only with a **certificate** attached.

- The certificate states: **device identity**, how the record was extracted, **who extracted it**, date and time
- It must carry the **SHA-256 hash** of the record - the technical proof of genuineness
- Complete this paperwork **during the investigation**. Making it later is difficult and looks suspicious
- Warning Without the certificate, digital evidence turns weak in court

Real Case: Sulli Deals and Bulli Bai

In 2021-22, women's photos were posted without consent on fake 'auction' websites. The investigation became a **textbook in evidence handling**.

- Police seized the accuseds' devices following **correct procedure** - sealed, tagged, chain of custody recorded
- Because the lifting was clean, the digital evidence **stood in court** despite aggressive defence challenges
- **Lesson:** however strong the technical evidence, wrong lifting makes the case weak. Right lifting makes it stand
- Source The Indian Express

Real Case: Cosmos Bank Attack, Pune

In 2018, malware let criminals withdraw **crores of rupees** from a Pune bank. The investigation taught a second lesson.

- Police seized laptops, phones and devices of the accused - the device side was handled well
- But equally critical were the **bank's server logs** - secured through legal process before deletion
- **Lesson:** in financial cybercrime, device seizure plus **immediate log preservation** - do both, fast
- Source The Indian Express

Your Turn: Labs in This Module

Evidence handling is a **hand skill**. Chapter 2 has four labs - do all of them this week.

- **Lab 1:** Hash experiment - change one character in a file and watch the entire hash change
- **Lab 2:** Mock crime scene - three-level photography plus a complete photo log
- **Lab 3:** Fill a full evidence tag - every field, no blanks, then photograph the tag
- **Lab 4:** FTK Imager - build a forensic image with write-protection and verify the hash

Key Takeaways

- **Think before touching** - digital evidence is fragile; one click can destroy it
- Five stages, in order: **identify, secure, record, pack, chain of custody**
- **Hash is the seal** - note it at seizure, during imaging, and before court
- Never examine the **original** - all analysis happens on the forensic copy
- Volatile evidence **first** - RAM and live data die when power goes
- Complete the **BSA certificate** during investigation, not after

Review Questions

- 1. An officer switches on a seized phone 'just to check one message'. List three things that change, and explain how the defence will use each in court.
- 2. You find a running laptop at the scene. Walk through your first five actions in order - and justify each one.
- 3. The chain-of-custody register has a two-week gap. As the defence lawyer, write the exact question you would ask. As the IO, how would you have prevented it?
- 4. Why must the hash be noted at three different points instead of just once?
- 5. A screenshot of a WhatsApp chat is submitted without URL, date or context. Argue for the prosecution: how do you strengthen it?

NEXT: MODULE 3

Evidence & Information Gathering

OSINT, email headers, WHOIS, verification & reporting